

Association of Efimov trimers from three-atoms continuum

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We develop an experimental technique for rf-association of Efimov trimers from three-atoms continuum. We apply it to probe the lowest Efimov energy level in bosonic lithium in the region where strong deviations from the universal behavior are expected, and provide quantitative study of this effect. Position of the Efimov resonance at the atom-dimer threshold, measured with a different experimental technique, concurs with the rf-association results.

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Low energy few-body physics with resonantly enhanced two-body interactions is a fundamental quantum mechanical problem, known to manifest universal properties [1]. For three interacting particles universality is associated with a ladder of weakly bound triatomic states called Efimov trimers that exist even when the two-body interactions support none [2]. Recent experiments with ultracold atoms have been highly successful in demonstrating several aspects of universality [3–18].

In the limit of zero collision energy, the two-body interactions are determined by a single parameter, the s-wave scattering length a . The window of universality is opened when a greatly exceeds the characteristic range of the interatomic potentials, the van der Waals length r_{vdW} . Analytical expressions for three-body experimental observables are derived in this window as the leading order expansion in powers of r_{vdW}/a . Their log-periodic dependence on a reveals the discrete scaling invariance of the underlying Efimov physics. The impact of finite range of interaction potentials on the lowest Efimov energy level in the context of ultracold atoms has been considered in recent publications [19–23],[34] and its role in the ongoing experiments is continually debated [4–6, 21–23]. Most straightforwardly, corrections in the next-to-leading order can be taken into account via the effective range of the interatomic potential R_e which is extracted from the s-wave scattering phase shift $\delta(k)$ at small relative wavenumbers k using the effective range expansion $k \cot \delta(k) = -1/a + R_e k^2/2$. However, it was noted that in this approach other effects of the same order have to be included through a new model-dependent parameter [22, 24].

Efimov physics in ultracold atoms is studied experimentally in the vicinity of Feshbach resonances which allow tuning of a via external magnetic field bias [25]. The usual three-body observable which shows log-periodic dependence on a constitutes the three-body recombination loss rate (K_3) of atoms from a shallow optical trap [3–16]. This experimental approach is only sensitive to a few points associated with each Efimov energy level. For negative scattering length enhancements in K_3 for certain values of a probe Efimov resonances which occur when the trimer states intersect with the three-atoms contin-

uum threshold [3]. For positive scattering length the two lowest Efimov potentials open two pathways to a deeply bound dimer which interfere destructively for certain values of a providing appearance of recombination minima in K_3 spectrum [3, 5]. In addition, signatures of Efimov resonances at the atom-dimer threshold can be observed through secondary collision processes [5].

Here we report on a different experimental technique in which we directly associate Efimov trimers from the three-atoms continuum. This approach applies a strong in-resonance rf-modulation of the magnetic field to stimulate free-atoms to trimer transition in the vicinity of the Efimov resonance at the atom-dimer threshold. At this limit the trimer is strongly asymmetric and resembles a “composite dimer” which consists of a dimer and a nearly free atom. Fig. 1(a) shows how close the dimer and the trimer energy levels are expected to be in this region. Near the threshold point the atom-dimer scattering length is resonantly enhanced [1] and can largely exceed a which shapes the trimer wavefunction with favorable spatial extension to overlap with the free three-atoms scattering state. High rates of dimer association were recently observed in our system in the relevant range of a [15, 26]. Here we extend this approach and study the formation of trimers. Note that their lifetime is expected to be short because of either a dissociation into high kinetic energy constituents (a deeply bound dimer and an atom) or an atom-trimer inelastic collision. Thus, trimers are quickly lost from the trap and their binding energy is detected as a loss resonance in rf-frequency scan measurements. This method provides, as a function of a , a continuous spectroscopic probe of an Efimov energy level and reveals its clear deviation from the predictions of universal theory which we attribute to the manifestation of range corrections. The general trend of our data consents with the shifted position of the Efimov resonance at the atom-dimer threshold that we detect through the secondary collision resonance in the three-body recombination rates measurements. Unfortunately, only partial theory is currently available for bosonic lithium which predicts a single point in the energy spectrum - the position of the Efimov resonance at the atom-dimer threshold [22]. This prediction is in a very good agreement with

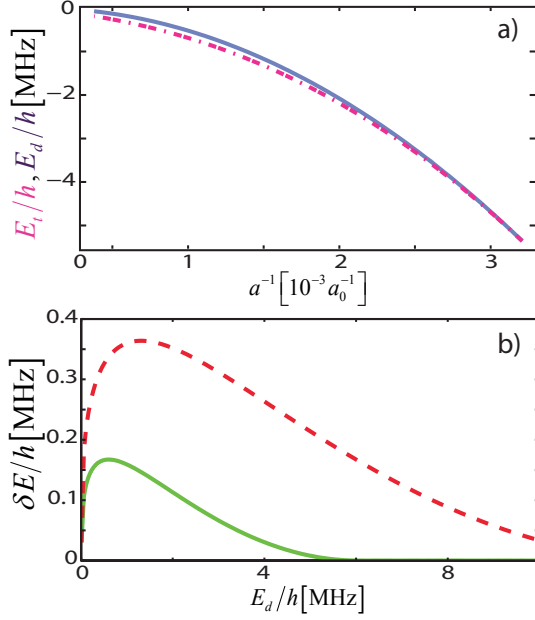


FIG. 1: (a) Dimer (solid purple) and trimer (dotted-dashed pink) energy levels follow each other very closely as calculated from universal theory [1] with $a_* = 295a_0$, where a_0 is the Bohr radius. This value is expected from the previously measured three-body recombination spectrum [15, 26]. (b) Difference between the dimer and the trimer energy levels for two different Efimov resonances at the atom-dimer threshold: $a_* = 295a_0$ (solid green) and $a_* = 200a_0$ (dashed red).

our results.

A similar experimental approach has been recently explored in fermionic ^6Li atoms [17, 18]. We hereby note two important differences. Primarily, extreme collisional stability of weakly bound dimers formed by two fermions [27] allows rf-association of trimers from the atom-dimer continuum [17]. This property, being the result of Pauli exclusion principle, is not applicable for bosonic ^7Li and makes the implementation of this experimental approach challenging. We thus explore direct association of trimers from the three-atoms continuum which is more general and applicable for all particles independent of their quantum statistical nature. Secondly, the two isotopes have very different Feshbach resonances [25]. The resonances in ^6Li are strongly entrance-channel dominated and thus characterized by large values of the resonance strength parameters $s_{res} \gg 1$ corresponding to hundreds of Gauss extension of the magnetic field range where $R_e \ll r_{vdW}$. In contrary, the resonances in ^7Li , while being almost as broad as their ^6Li counterparts, have an intermediate character between entrance-channel and close-channel dominated with $s_{res} \sim 1$. This difference, caused by the anomalously large background scattering length in ^6Li , introduces the possibility to study consequences of early breakdown of universality in two-body physics on deviations from the universal be-

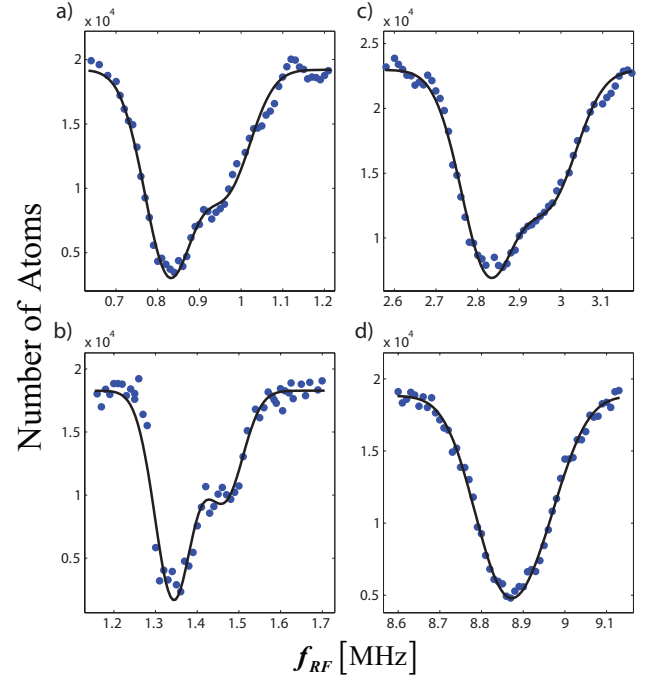


FIG. 2: Rf-induced atom loss resonances are shown for different values of bias magnetic fields corresponding to different values of dimer binding energies. For weakly bound dimers ((a)-(c)) strongly overlapped double-peak loss structures are clearly visible. For deeper bound dimers (d) only a single minimum is observed signifying that the dimer and the trimer energy levels converged to a value below the resolution limit. Each point is an average of 3 to 5 individual measurements. Solid lines represent the double-peak Gaussian fits to the data.

havior in three-body physics.

The experiments are performed on a gas of ^7Li atoms polarized on the $|F = 1, m_F = 0\rangle$ state and evaporatively cooled down to a typical temperature of $\sim 1.5\mu\text{K}$ in an optical trap [28]. In earlier studies of the a -dependent spectrum of K_3 on this state in the vicinity of a broad Feshbach resonance at $\sim 894\text{G}$ we identified a recombination minimum at $a_0^* = 1130(120)a_0$ and a maximum at $a_- = -280(12)a_0$, where a_0 denotes the Bohr radius [7, 26]. The ratio of these two values is close to the universally predicted value [1] of $a_-/a_0^* = -\exp(-\pi/2s_0) = -0.21$, where $s_0 = 1.00624$, indicating that the three-body parameter is preserved across the Feshbach resonance. Using universal relations we can predict the position of the Efimov resonance at the atom-dimer threshold at either $a_* \approx 1.1 \cdot a_0^*/\sqrt{22.7} = 262(27)a_0$ or $a_* \approx -1.03a_- = 288(12)a_0$ using the recombination minimum or maximum respectively [1].

We apply a strong rf-modulation at a given magnetic field and scan its frequency to obtain atom loss resonances. Modulation amplitude induced by the rf-field reaches 1.6 G at full rf-power which allows for $\sim 40\%$ atom-dimer conversion efficiency within a short time of 1 msec at a typical mean density of $\sim 10^{12}\text{ cm}^{-3}$ and

$a \cong 1000a_0$. While the rate of molecule association is defined by the two-body collision rate ($\sim 3\text{kHz}$ for the above parameters), the trimer association rate relies on significantly rarer three-body collisions which require three atoms to be at the same position in space. However, we do not measure the three-body collision rate, but rather the three-body recombination-induced loss of atoms from the trap. In case of dimer association, quick loss is observed when an associated dimer occasionally finds a collision partner and decays to a deeper bound state releasing enough kinetic energy to cause the loss of all colliding parties. By addressing directly the Efimov energy level we bind together all three collision partners, initiating a fast and efficient decay process. Consequently, inelastic losses are resonantly enhanced, and we can make use of this enhancement by matching each magnetic field value with rf-modulation time, which is kept to be an order of magnitude less than the corresponding lifetime of atoms due to three-body recombination. The latter is defined as $1/\tau = K_3\langle n^2 \rangle$, where $\langle n^2 \rangle$ is the mean of squared atomic density [7]. The rf-modulation time varies accordingly, between $\sim 50 - 2000$ msec for different magnetic fields. After switching off the rf-field we detect the remaining number of atoms in the trap. A few rf-scans for different values of magnetic field are shown in Fig. 2(a)-(d). A typical signal represents strongly overlapping but clearly visible double-peak loss structure in which the lower rf-frequency and deeper loss minimum corresponds to the well known dimer association resonance, which we used earlier for precise characterization of the Feshbach resonance [26]. However, this signal is now power-broadened due to the high rf-power. We associate the second loss minimum with the formation of Efimov trimers. For low energy dimers, the trimer signal is clearly visible (Fig. 2(a)-(c)). However, for deeper bound dimers it disappears (Fig. 2(d)), as expected, when the trimer and dimer energy levels' separation are less than the typical width of the power-broadened dimer resonance ($\sim 110\text{kHz}$). Narrower dimer resonances were observed for lower rf-powers and shorter rf-modulation times but with the expense of losing the trimer signal.

It is interesting to note that the rf-scans in Fig. 2 are well fitted with two overlapping Gaussian distributions of the same widths. Most probably it indicates that the trimer resonance width is not defined by the finite lifetime of trimers which then should exceed $10\mu\text{s}$. On the other hand, trimers do not live longer than the inverse of the two-body collision rate which limits their lifetime from above to few hundreds of μs for typical phase space densities used in the experiment. This range opens an experimentally realistic time slot to manipulate the trimer state. However, direct rf-association of trimers discussed here limits their production rate to the rate of three-body collisions which is significantly slower than their decay rate that relies on two-body collisions. This might pre-

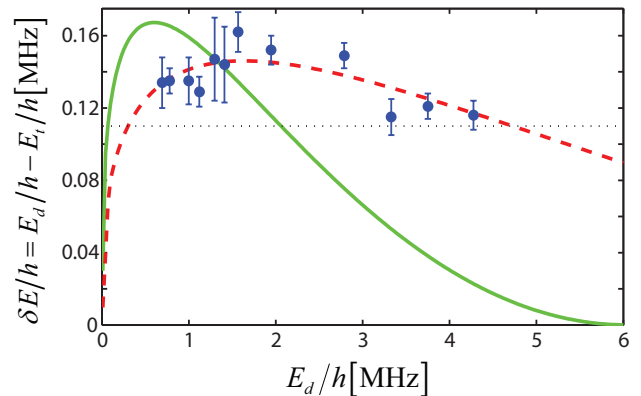


FIG. 3: Difference between trimer and dimer energies as a function of the dimer energy level. Points represent results of fits to the measurements with double-peak Gaussian distributions. The error bars show one standard deviation of the fit. Solid line represent prediction of universal theory for $a_* = 295a_0$. Dashed red line shows fit of universal theory with an additional fitting parameter - the multiplication factor of the energy difference δE . Dotted horizontal line marks the minimal δE that can be reliably detected due to the resolution limit.

vent the observation of macroscopic ensemble of Efimov trimers in current approach.

We fit the rf-scans with a double-peak Gaussian distribution to determine the position of the dimer resonance, its width and the difference between the dimer and trimer energies $\delta E = E_d - E_t$. In Fig. 3, δE is shown as a function of the dimer energy E_d . By representing only δE we eliminate the common shift of the energy levels due to finite temperature and strong magnetic field modulation. The points in Fig. 3 represent measurements while the green solid line shows prediction of universal theory with $a_* = 295a_0$ which is within the error of the values derived from the $K_3(a)$ measurements. At first glance a clear disagreement between measurements and theory is observed. Universal theory predicts that the trimer and the dimer energy levels will converge sooner than obtained experimentally. For example, for $E_d \approx h \times 2.8\text{MHz}$ δE should be well below the resolution limit, preventing the observation of the trimer signal, but it is clearly visible and disappears for significantly larger E_d (see Fig. 2(c),(d)). Note however, that universal theory predicts the depth of the trimer state δE reasonably well leaving the main disagreement between theory and experimental results in the convergence of the two energy levels. Our measurements then suggest that the Efimov resonance at the atom-dimer threshold is shifted to a lower scattering length value, but they also suggest that theory should go beyond universality to correctly describe our data. If we were using a lower a_* value within the framework of universal theory we would obtain larger δE in the relevant range of E_d in clear contradiction to our results (see Fig. 1(b)). Theory of full Efimov en-

ergy spectrum with finite range corrections for bosonic lithium does not yet exist. However, recently an effective field theory approach has been used [22] to calculate the shift of a_* to a new position a_{*e} based on $R_e(a)$ and $K_3(a)$ that were reported in our previous work [7]. This “beyond universality” theory predicts $a_{*e} = 210(44)a_0$ in agreement with the general trend of the experimental results. With the absence of full theory and in order to estimate a_{*e} with which the measurements represented in Fig. 3 will agree, we modify the universal theory by introducing an additional parameter, a multiplication factor of the energy difference δE . We fit our data with this model and the result is shown in Fig. 3 as a dashed red line which describes our measurements reasonably well. It estimates $a_{*e} \approx 180a_0$ which agrees with the available “beyond universality” theory result. Of course fitting of our data with a full theory is ultimately needed to correctly predict the value of a_{*e} .

Independently, a_{*e} can be detected via secondary collisions resonance expected to appear in $K_3(a)$ spectrum [5]. A secondary collisions resonance occurs when a dimer with relatively high kinetic energy is formed in a three-body recombination process and collides with other atoms on its way out of the trap, resulting in loss of more than three atoms in a single three-body recombination event. Such a resonance is likely to occur in the vicinity of the Efimov resonance at the atom-dimer threshold where atom-dimer elastic and inelastic collisional cross-sections are enhanced by orders of magnitudes [1]. Here we reveal $K_3(a)$ measurement that we earlier performed on the absolute ground state ($|F = 1, m_F = 1\rangle$) [15, 26]. Fig. 4 shows $K_3(a)$ for positive scattering lengths in the vicinity of the Feshbach resonance at $\sim 738\text{G}$. In addition to the already indicated three-body recombination minimum at $a_0^* = 1260(76)a_0$, a narrow peak appears at $\sim 200a_0$. We can now attribute this peak to a secondary collision resonance that reveals the position of a_{*e} . To make a quantitative argument in favor of this attribution we developed a model which counts the elastic and inelastic collisions of a dimer with trapped atoms based on the available analytical expressions for the cross-sections of these events [29]. Thus the model predicts enhancement of atom loss per three-body recombination event. We fit our data with this model using three fitting parameters: height of the peak (which has no meaning as we do not model the K_3 coefficient directly), a_{*e} and η which describes the lifetime of a trimer. The fit, shown as a solid line in the inset of Fig. 4, demonstrates excellent agreement with the measurement which strongly supports our treatment of the feature as an avalanche resonance. We note that the fitting parameter $\eta = 0.072(1)$ is smaller by a factor of ~ 2.5 compared to the experimentally observed values for minimum and maximum in the $K_3(a)$ spectrum [15, 26]. This disagreement might be attributed to yet another manifestation of “beyond universality” corrections required to explain our experi-

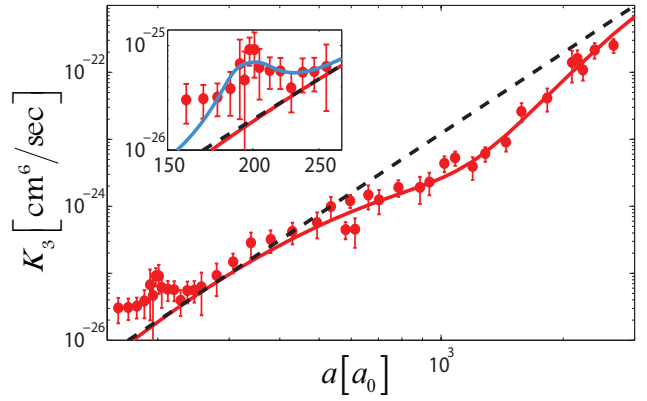


FIG. 4: Three-body recombination spectrum. Only positive range of a is shown. In addition to the earlier published minimum in $K_3(a)$ [15] there is a local maximum (enlarged in the inset) that we attribute to a secondary collisions resonance which indicates the position of the Efimov resonance at the atom-dimer threshold $a_e^* \approx 200a_0$. The solid line in the inset is a fit to a model of secondary collisions [29].

mental results.

The fitted position of $a_{*e} = 196(4)a_0$ does not agree with the universal theory prediction of either $a_* = 292(18)a_0$ or $a_* = 282(12)a_0$ using a_0^* or a_- , respectively. Although these measurements are performed on the absolute ground state instead of the one but lowest Zeeman sublevel, under discussion here, we showed earlier that the three-body parameter is the same for both states within the experimental errors [15, 26]. As the effective range R_e is also similar [7, 22], we expect nearly the same position for a_{*e} which is, then, in good agreement with both “beyond universality” theory [22] and the above estimation from the direct measurement of the Efimov energy level. We note that our original $K_3(a)$ measurements on $|F = 1, m_F = 0\rangle$ level [7] were not dense enough in the relevant region of a to identify a similar avalanche resonance and they have to be revisited to reveal this feature.

It is interesting to compare our results with those obtained for ^6Li , especially in view of the experimentally discovered universality in the position of the Efimov resonance at the free atoms threshold (a_-) in the region of $a < 0$. In all but one atomic species $a_-/r_{\text{vdW}} \sim -9.5$ within $\pm 15\%$ [5–7, 10–13, 15, 16, 30] which has been recently explained in a landmark theoretical work [31] [35]. Despite this universality in a_- , a_{*e} varies significantly between different species [4, 5, 13, 14, 29] which is a clear consequence of corrections caused by the finite range of interatomic potentials. However, r_{vdW} for both isotopes of Li is very similar [25], still a_{*e} differs significantly. In our case $a_*/a_{*e} \sim 1.5$, but for ^6Li corrections go in the opposite direction, i.e. $a_*/a_{*e} < 1$ [13, 14]. This emphasizes the role plying by s_{res} parameter in the breakdown of three-body universality in ^7Li .

In conclusion, we develop a general experimental technique for rf-association of Efimov trimers from three-atoms continuum and probe with it the lowest Efimov energy level. We identify clear deviations of the energy level from universal theory which we attribute to the manifestation of finite range corrections. We show that the position of the Efimov resonance at the atom-dimer threshold, measured via a secondary collisions resonance in three-body recombination spectrum, concurs with the rf-association results. Our experimental method paves the way to direct population, manipulation and interrogation of Efimov quantum states. During short but experimentally reasonable lifetime of Efimov trimers, they can be transferred to a deeply bound state and their macroscopic ensemble can then be realized. We note also that a recently developed theory has found reasonable rf association rates of Efimov trimers in bosonic lithium for similar phase space densities [32]. However this theory uses a different starting configuration (one boson is distinguishable) and thus cannot be directly applied to our system.

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